

Reward and Controller Comparisons for Cellular UAVs under Shared Connectivity Constraints

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Abstract—We compare a master’s thesis reward with a replacement within a shared cellular UAV control system, V2, using the same Barcelona ray tracing dataset as the original thesis. The original reward arm, V1.5, shares V2’s dynamics, observations, navigation prior, safety filter, PPO settings and strict sampled connectivity rules. Five training seeds and 400 fresh shared routes support the initial comparison. Full V2 achieves 95.7% standard and 87.3% longer joint success, versus 95.7% and 85.8% for V1.5. The primary standard difference is +0.0 percentage points, with a 95% crossed seed and route bootstrap interval $[-2.8, 2.7]$. The primary comparison does not establish improved completion from reward replacement. Full trades fewer handovers for higher delay. Every successful mission meets the sampled RSS and buffer constraints. These results concern our declared simulator; the original implementation remains unavailable. A subsequent validation selected logarithmic delay reward was tested on 1,000 new shared routes with five seeds. Its standard completion difference versus full V2 is -1.440 points $[-3.160, 0.320]$. The fresh standard comparison does not establish improved completion over full V2. A separate fixed weight lookahead supervisor has a fresh standard completion difference of +2,000 points $[0.320, 3.681]$ versus full V2. The primary fresh comparison supports improved standard mission completion. The further signal path rule has mean SNR differences of +0.377/+0.284 dB versus V2.2 on 10000 fresh episodes, and misses its declared combined gate.

Index Terms—Cellular UAV, connectivity constraints, reward design, handover, trajectory control, resource allocation, reinforcement learning, reproducibility.

I. INTRODUCTION

A master’s thesis studies joint trajectory, cell association and resource control using Barcelona ray tracing data [1, pp. 6–7, Sec. 3.1]. Its objective weights delay, interference and handovers [1, p. 10, Eq. (11)], with minimum serving RSS and bounded queue constraints [1, p. 11, Eq. (12), C2 and C4]. The thesis reports wandering and prioritization of connectivity over destination completion [1, p. 19, Sec. 7.2; p. 21, Sec. 8]. Positive reciprocal radio utilities and a fixed approach bonus [1, p. 12, Eqs. (13)–(14)] motivate an empirical reward audit, but do not establish the cause of behavior in unavailable original code.

The central question is whether replacing that reward improves joint mission success when every other component is the same V2 system. V1.5 denotes the written original reward on V2, created as a new controlled intervention. Both arms retain V2’s strict sampled connectivity rules, arrival termination, navigation prior and prospective safety filter. We do not restore the original continued training after arrival [1,

p. 12, Sec. 5.2]. The experiment therefore measures the reward package conditional on V2, not the complete original agent or individual reward terms in isolation.

We contribute a transparent reward comparison, fresh matched tests, all training seeds, explicit failures and reproducible artifacts. PPO [2] is an established algorithm; we claim no new learning method. Thesis citations give printed pages; add two for the PDF viewer page. We use the same thesis dataset; the original simulator and trained policy remain unavailable. We also recover source motivated metrics and test a separately declared lookahead supervisor using unchanged full V2 weights on a third fresh route set. We then test a separately frozen signal aware path selection rule against that supervisor on a fourth fresh route set, retaining unsuccessful candidates and the declared completion and service gates.

II. SOURCE REQUIREMENTS

A. Which Communication Requirements Apply?

We use the thesis requirements, not a newly imposed application service level. The RSS minimum is -96 dBm and the buffer is listed as 160 KB [1, p. 15, Table 3]; we retain the explicit 1,280,000 bit interpretation. The written C2 inequality allows equality and applies at every modeled time [1, p. 11, Eq. (12)]. Delay remains an optimized quantity, defined in the source as queue divided by service rate [1, p. 9, Eq. (4)]. Neither a two second packet deadline nor an empty queue arrival condition is introduced. A successful arrival can therefore retain queued data; it must not be described as completed data delivery.

The distinction between formulation and enforcement matters. The source reward penalizes RSS violations [1, p. 12, Eq. (14)], which does not establish that its unavailable implementation guaranteed C2. Its direct data dump penalty was removed during development [1, pp. 20–21, Sec. 7.3]. We compare verified written definitions with our executable rules, not inferred behavior of missing original code.

III. SHARED V2 SYSTEM

A. Shared Physical Assumptions

We use the same Barcelona ray tracing dataset as the original thesis, preserving its native resolution RSS map and selecting the first operator’s 87 stations. The dataset represents 2.1 GHz and 100 m altitude [1, p. 7, Table 2; p. 15, Table 3]. Its identity was confirmed by the researcher who supplied

it. The raw private HDF5 is not modified or published. The provisional -128 sentinel interpretation remains unverified.

The shared model retains a one second decision step, acceleration norm limit 5 m/s^2 , speed limit 25 m/s , 200 s deadline, and a 100 kJ budget under the engineering energy proxy $100 + 0.4v^2$ watts. These are shared V2 settings. They differ from the source's 0.1 s table step and 100 m/s speed setting [1, p. 15, Table 3]; the source also gives a conflicting 1 ms interval in prose [1, p. 14, Sec. 6.1].

The radio proxy retains 12 resource groups, half occupied by a deterministic station pattern. Interference sums linear received power from cochannel background stations. Capacity is $1.44 \cdot 10^6 \log_2(1 + \text{SINR})$ bit/s, with noise -103.41 dBm and offered traffic $200,000 \text{ bit/s}$. The source's printed rate and SNR expressions differ [1, p. 10, Eqs. (7)–(8)], and its background occupancy is described as random [1, p. 14, Sec. 6.1]. We retain these declared surrogate assumptions to separate the current repair from a new propagation or traffic model. The interference is a downlink proxy, not a calibrated uplink quantity matching the source's terminology [1, p. 10, Eq. (9)].

The shared motion preserves inertial velocity and uses trapezoidal position integration, unlike the source scalar speed/direction update [1, p. 9, Eq. (2)]. The queue receives constant traffic and executed handover overhead, subtracts this step's service, then records overflow before clipping. Delay is postservice backlog divided by a rate floor of one bit/s, censored at 200 s . The source supplies the queue/rate concept and packet formulation [1, pp. 9–10, Eqs. (4)–(6)]; our constant bit rate and service proxy do not reproduce individual packet latency.

A full grid scan finds at least 38 selected operator stations above -96 dBm at every location. Direct trajectories can therefore be appropriate; actual association restrictions and interference still matter. Curvature is not an objective. Buildings affect the inherited propagation map, but no obstacle collision model is implemented.

B. Joint Success and Failure

Let d_t denote destination distance and q_t the queue after the sampled arrival/service update. Joint success requires arrival by the deadline with $d_T \leq 10 \text{ m}$ and $\|v_T\| \leq 2 \text{ m/s}$, together with

$$\text{RSS}_{k_t}(p_t) \geq -96 \text{ dBm}, \quad 0 \leq q_t \leq 1,280,000 \text{ bits}, \quad \forall t. \quad (1)$$

The 10 m tolerance follows the source table [1, p. 15, Table 3]; the numeric stopping tolerance is our explicit interpretation of stopping at the destination [1, p. 9, Sec. 3.2]. Map and energy conditions also apply. A single sampled RSS violation or any buffer overflow ends a failed mission. Failure takes precedence over simultaneous arrival. Initial RSS is checked; queue clipping for storage cannot erase an overflow flag.

This enforces the constraints at modeled samples. It is not a guarantee between the one second samples or within the queue update. No minimum SINR, packet latency threshold, or final queue draining requirement is claimed.

C. Actions, Residual Motion and Safety Filter

The source includes BS and RBG actions [1, p. 12, Sec. 5.2]. We expose 61 categorical options: stay, or five candidate station slots by 12 RBGs. Candidates are serving plus the four strongest RSS stations. Masks reject occupied groups and duplicated serving station slots. A new station must have current RSS more than 3 dB above the serving RSS and meet the minimum, following the source's A3 direction [1, p. 9, Eq. (3); p. 15, Table 3]. Network decisions occur each modeled second. Resource changes within the same station are allowed and counted separately from handovers. Handovers generate $4,800$ control bits [1, p. 15, Table 3]; switching interruption is not modeled.

The observation has 156 features covering geometry, velocity, time, queue, energy, serving SINR, candidate RSS and positions, option capacities and masks. Two Gaussian outputs define bounded residual acceleration commands. We project a destination directed braking command onto the unit disk, add half of the tanh transformed residual, and apply the shared acceleration projection. All learned arms use this prior; they do not learn basic flight entirely from scratch. The source's fully discrete action and smaller state differ [1, p. 12, Sec. 5.2].

A shared safety filter predicts the proposed next position, RSS and queue. It first tries another admissible station/resource pair for the same motion. If none is feasible, it invokes the local motion/network search. If no candidate repairs the step, the flight fails normally. The filter has prospective access to the known map and is an explicit control component, not learned competence. Interventions are recorded. It cannot establish global reachability.

D. The V2 Full and Arrival Rewards

Using source equal weights and scales [1, p. 15, Table 3], define

$$C_t = 0.35 \frac{10D_t}{1 + 10D_t} + 0.30 \frac{10^5 I_t}{1 + 10^5 I_t} + 0.35 \frac{100H_t}{1 + 100H_t}. \quad (2)$$

Here D_t is the backlog/service delay proxy in seconds, I_t interference in watts, and H_t a handover indicator. These are one minus the source reciprocal utilities [1, p. 12, Eq. (13)], removing the positive reward for merely remaining active while retaining all communication components. The reciprocal cost still saturates smoothly, which can favor substantial delay in exchange for fewer handovers; it is not a latency guarantee.

The V2 full and arrival arms use

$$\Phi_t = -\frac{d_t + \|v_t\|^2 / (2a_{\max})}{100 \text{ m}}, \quad (3)$$

with terminal potential zero. The base reward is $0.99\Phi_{t+1} - \Phi_t - 0.05$, plus 20 for joint success and minus 20 for terminal failure. The arrival arm uses this base alone; the full arm subtracts C_t . Both have identical hard failure rules and control interfaces. This is not a comparison to the original trained thesis policy or a claim that a finite scalar reward proves optimal lexicographic control.

IV. ORIGINAL REWARD INTERVENTION AND EXPERIMENTAL PROTOCOL

A. V1.5: Original Written Reward on V2

For executed postfilter delay D , interference I and handover h , use

$$r_{\text{orig}} = \frac{0.35}{1 + 10D} + \frac{0.30}{1 + 10^5 I} + \frac{0.35}{1 + 100h} + 12\mathbb{I}[d_{t+1} < d_t] - 10b - 10o - 10s. \quad (4)$$

The source supplies the algebra, equal weights and coefficients [1, p. 12, Eqs. (13)–(14); p. 15, Table 3]. Here b is a boundary violation, o is $\text{RSS} \leq -96$ dBm, and s is speed ≥ 100 m/s. Energy exhaustion replaces the whole reward with -10 [1, p. 12, Sec. 5.2]. No new arrival, timeout or overflow payment is added. The source’s removed energy utility and data dump penalty remain absent [1, pp. 20–21, Sec. 7.3].

The reward retains its RSS equality penalty although the joint constraint permits equality. Its speed threshold remains 100 m/s, unreachable under the V2 25 m/s cap. Substituting 25 m/s would invent a penalty on routine cruising. The same V2 transition supplies radio inputs, physics, filter interventions and terminal outcomes to all arms. The wrapper changes only returned reward and accumulated raw return. Full reward replacement changes the entire package; the effects of shaping, terminal payments and removing positive per step utilities are not separately identified.

B. Shared Training and Fixed Checkpoints

PPO uses separate two layer 64 unit tanh actor and critic networks, 64 lanes, 128 step rollouts, four epochs, minibatches of 512, learning rate 3×10^{-4} , Adam epsilon 10^{-5} , discount 0.99, GAE 0.95, clip 0.2, entropy weight 0.005, value weight 0.5 and gradient norm limit 0.5. Two CPU threads are used. Running discounted return variance normalizes rewards, clipped to $[-10, 10]$. This shared processing responds to changed raw rewards; potential algebra alone does not establish invariance after these transforms. The source uses Stable Baselines PPO [1, p. 13, Sec. 5.3] and different training settings [1, pp. 15–16, Sec. 6.2 and Table 4].

All arms use the exact 4,096 V2 training routes spanning 200–1800 m, 64 standard validation routes and 32 longer validation routes. Five initialization seeds, 2101–2105, and 524,288 interactions per policy are shared. Five original reward policies are newly trained from scratch: 2,621,440 new interactions. All ten completed full and arrival V2 policies are reused; they used identical training data, seeds and budget. Their hashes were fixed before new training and test evaluation. No seed is selected for its performance. Different policies can reset at different times, so equal seeds and budgets do not imply identical realized transitions. Every run uses its final budget checkpoint.

Earlier V2 test outcomes were known. This protocol was frozen before the new training and fresh test evaluation; it is not independently registered or claimed to predate the earlier V2 study. No new hyperparameter tuning or policy selection uses the fresh tests.

C. Fresh Tests and Prespecified Analysis

Generate 200 standard routes of 200–1000 m using seed 53012 and 200 longer routes of 1000–1800 m using seed 53013. Exact route pairs are disjoint from all prior saved routes; every controller shares initial conditions and load phases. Both lengths occur in training. This is route holdout on one map, not geographic generalization or length extrapolation. The source does not document equivalent independent splits [1, pp. 15–16, Secs. 6.2–6.3].

The primary contrast is full minus original standard joint success. Use 5,000 crossed seed and route bootstrap draws, seed 63000, to estimate a 95% interval. An interval entirely above zero supports improved primary completion under this design. Longer completion is prespecified secondary. Five seeds on 200 shared routes are not 1,000 independent training replications. Secondary intervals are descriptive and unadjusted for multiple comparisons.

Secondary metrics use the common successful seed/route pairs, with their count visible. Unconditional success and failure rates remain separate. No percentage is computed for logarithmic SINR. Three learned arms supply 6,000 episodes; four deterministic references supply 1,600, totaling 7,600 new final evaluations. These records are separate from the previously evaluated V2 routes.

D. Deterministic References and Verification

Goal RSS uses nominal braking motion and strongest admissible association. Goal radio uses the same motion and prospective normalized radio cost selection. One step and three step joint search examine nine initial motion primitives, varying direction and speed. The three step version follows each initial choice with two predicted reference steps. All share the same safety filter, which can alter nominally straight paths. These local methods can fail and are not a reproduction of the source greedy controller [1, pp. 13–14, Sec. 5.4].

All 73 tests pass, including unchanged states, observations, masks and terminal records under identical actions across reward arms. Failure checks cover RSS, overflow, boundary, energy and timeout as well as arrival. A syntax comparison verifies that copied PPO differs only in its environment import and module description. Small full reward training reproduces frozen V2 parameters exactly; original reward training repeats exactly in its deterministic check. All 7,600 new final episode records replayed exactly. Source, dataset and checkpoint hashes remain unchanged. These checks verify software reproduction, not physical model validity.

V. RESULTS ON FRESH SHARED ROUTES

A. Primary Reward Contrast

Full achieves 95.7% standard success versus 95.7% for V1.5. The paired difference is +0.0 percentage points, 95% interval $[-2.8, 2.7]$. The primary comparison does not establish improved completion from reward replacement. An interval containing zero is not proof of equivalence. On longer routes the difference is +1.5 points, interval $[-2.9, 6.1]$. The longer route interval also includes zero; its observed difference does

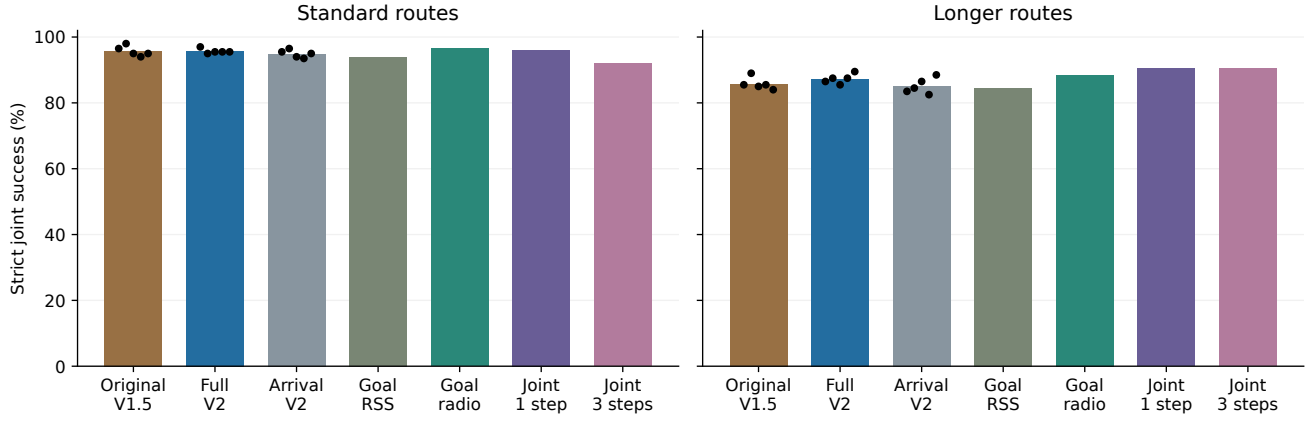


Fig. 1. Joint success on fresh shared routes. Bars aggregate episodes; dots show individual learned training seeds, not confidence intervals. Deterministic controllers have one evaluation per route.

TABLE I
STRICT JOINT SUCCESS ON THE NEW SHARED ROUTES (PERCENT).

Controller	Standard	Longer	Per split
PPO original (V1.5)	95.7	85.8	1,000
PPO full (V2)	95.7	87.3	1,000
PPO arrival (V2)	94.9	85.1	1,000
Goal RSS	94.0	84.5	200
Goal radio	96.5	88.5	200
Joint one step	96.0	90.5	200
Joint three steps	92.0	90.5	200

TABLE II
FIRST FAILURES: RSS / BUFFER / TIMEOUT COUNTS. NO ENERGY OR BOUNDARY FAILURES OCCURRED.

Controller	Standard	Longer
PPO original (V1.5)	40 / 3 / 0	136 / 6 / 0
PPO full (V2)	34 / 9 / 0	112 / 15 / 0
PPO arrival (V2)	51 / 0 / 0	147 / 0 / 2
Goal RSS	12 / 0 / 0	31 / 0 / 0
Goal radio	7 / 0 / 0	21 / 2 / 0
Joint one step	8 / 0 / 0	18 / 0 / 1
Joint three steps	4 / 0 / 12	11 / 0 / 8

not establish improvement. Every training seed appears in Table IX.

B. Failures and Communication Tradeoffs

Table II reports first failure reasons. Episodes stop on the first violation; hypothetical later failures are unknown. Every successful flight has zero sampled RSS outage and zero dropped bits.

Full versus original secondary comparisons contain 925 common successful standard seed/route pairs and 781 longer pairs. Table III reports conditional paired values and intervals. Higher success and SINR are preferable. Lower delay, interference, consumed energy and flight time are preferable on common successful missions; fewer handovers are useful only

while preserving service. A lower weighted radio cost does not imply improvement in every component. On standard common successes, accumulated radio cost changes by -25.2%, while delay changes from 1.94 to 5.83 s. Flight time changes by -3.14 s and handovers by -6.40. On longer common successes, delay changes from 2.86 to 6.82 s. Interference differences remain inconclusive on both splits. A favorable aggregate cost is not interchangeable with delay, interference, handovers or completion. Different successful subsets must not be treated as matched counterfactual outcomes.

C. Other Controls

Full versus arrival completion differences are +0.8 points [-1.8, 3.4] on standard routes and +2.2 points [-2.3, 6.7] on longer routes. Both arms enforce strict constraints. Goal radio attains 96.5% and 88.5% joint success. Joint search has the highest observed longer completion on this route set, while Goal radio has the highest standard completion. These observed rankings are not a demonstrated overall superiority claim. These references expose performance available from the shared model rather than demonstrating learned superiority.

D. Illustrative Long Route

Figures 2–3 show route 53013_0023, the longest Euclidean start to goal distance among all 200 declared longer test routes (1,784.7 m). Ties use scenario ID; learned seed 2101 is retained. Selection uses geometry only. Dense replay of the three original 200 route batches matched all 600 records exactly; this illustration adds no independent observations or training.

All three selected flights succeed: original reward takes 85 s and 18 handovers, full takes 78 s and one handover, and Goal radio takes 77 s and two handovers. Mean delay proxies are respectively 0.78, 2.79 and 0.87 s. Full's queue approaches capacity while RSS stays above the minimum. Paths largely overlap under the shared navigation aid, despite different service behavior. Full's accumulated radio cost is higher than original on this route, unlike the average contrast. Aggregate results determine the conclusions.

TABLE III
FULL MINUS ORIGINAL ON 925 STANDARD AND 781 LONGER COMMON SUCCESSFUL PAIRS. SECONDARY INTERVALS ARE DESCRIPTIVE AND UNADJUSTED.

Split	Metric	Original	Full	Difference	95% interval
Standard	Flight time (s)	31.252	28.112	-3.139	$[-3.509, -2.808]$
Standard	Accumulated radio cost	7.094	5.309	-1.785	$[-2.194, -1.389]$
Standard	Delay proxy (s)	1.942	5.827	+3.886	$[2.731, 5.066]$
Standard	Handovers	6.843	0.443	-6.400	$[-8.501, -4.366]$
Standard	Interference (μW)	0.753	0.748	-0.004	$[-0.017, 0.009]$
Standard	Energy proxy (kJ)	7.651	7.788	+0.137	$[0.115, 0.157]$
Standard	SINR (dB)	-8.322	-9.305	-0.983	$[-1.483, -0.488]$
Standard	Path length (m)	547.596	551.204	+3.608	$[2.947, 4.273]$
Standard	Network overrides	0.107	0.478	+0.371	$[0.243, 0.504]$
Standard	Motion overrides	0.008	0.016	+0.009	$[-0.004, 0.028]$
Longer	Flight time (s)	65.394	60.145	-5.250	$[-5.815, -4.804]$
Longer	Accumulated radio cost	16.800	13.803	-2.997	$[-3.765, -2.257]$
Longer	Delay proxy (s)	2.857	6.818	+3.961	$[2.657, 5.145]$
Longer	Handovers	15.826	1.528	-14.298	$[-19.543, -9.101]$
Longer	Interference (μW)	0.794	0.791	-0.003	$[-0.015, 0.009]$
Longer	Energy proxy (kJ)	18.523	18.957	+0.434	$[0.394, 0.476]$
Longer	SINR (dB)	-8.773	-9.957	-1.184	$[-1.652, -0.712]$
Longer	Path length (m)	1345.958	1349.568	+3.610	$[2.896, 4.321]$
Longer	Network overrides	0.467	1.757	+1.289	$[0.882, 1.637]$
Longer	Motion overrides	0.024	0.044	+0.019	$[-0.010, 0.062]$

Longest declared test route: 1,784.7 m | Same start, goal and traffic

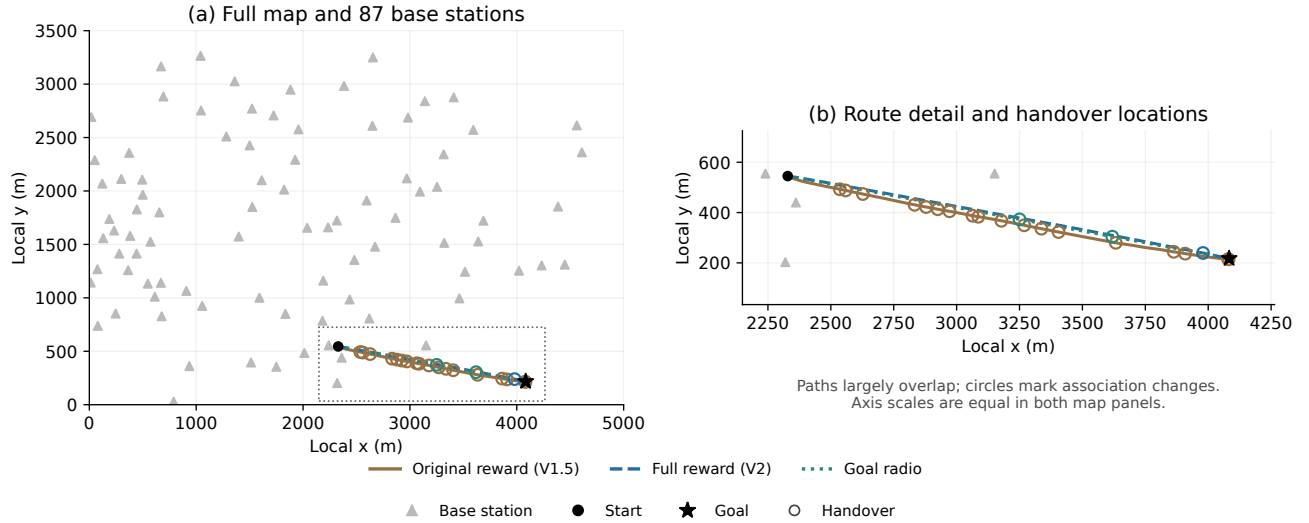


Fig. 2. Longest declared longer test route, with the full station map and route detail at equal coordinate scales. Triangles are the 87 Operator 1 base stations; colored circles mark executed handovers. Original reward, full reward and Goal radio share start, goal and traffic. All three succeed. Nearly overlapping paths do not imply identical communication service.

VI. COMPLETION FIRST SERVICE REWARD FOLLOWUP

After the initial comparison, we declared a separate V2.1 study to prioritize completion, then service. The full V2 delay cost is 0.318 at 1 s, 0.344 at 6 s and 0.347 at 10 s; a handover costs about 0.347. This saturation motivates a different reward, not a conclusion about trained performance.

A. Development and Frozen Protocol

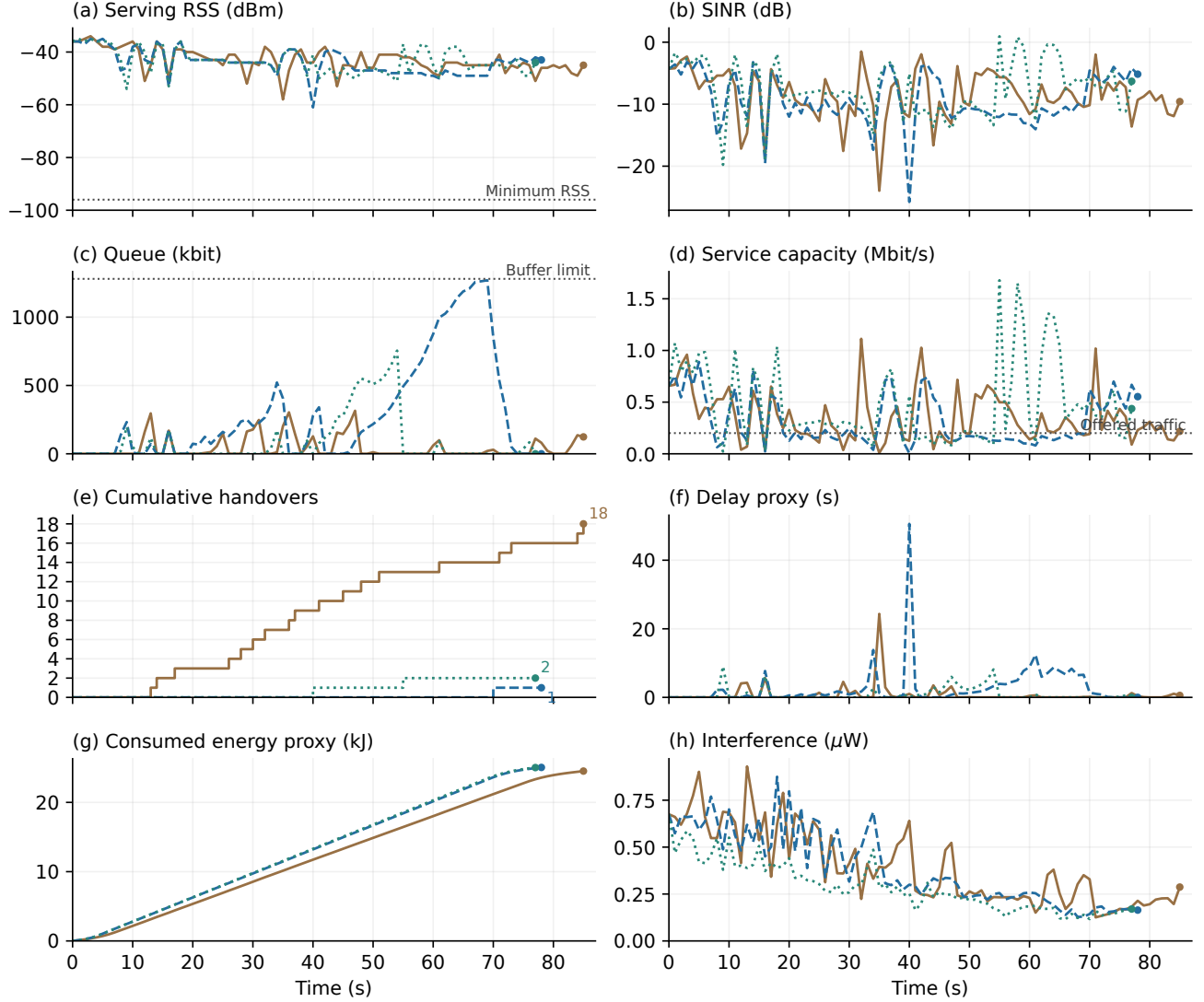
Four pilots used seed 2199 and 524,288 interactions each, with the same 64 standard and 32 longer validation routes. Full

V2 completed 90/96 missions, the service candidate 91/96, and two candidates with stronger RSS, queue and failure penalties 87/96 and 89/96. The declared rule selected by completion first, then matched delay and handovers; advancement required no larger matched delay in either split. All candidates were retained.

The selected service cost is

Communication and energy along the same 1,784.7 m route

— Original reward (V1.5) — Full reward (V2) ... Goal radio



Initial and executed samples at 1 s intervals; each curve stops at its episode endpoint.

Fig. 3. Communication and energy on the same route and controller runs as Fig. 2. Curves include the initial state and each executed one second sample, stopping at their own episode endpoints. Dotted horizontal lines show minimum RSS, buffer capacity and offered traffic. Capacity is the model's potential service rate, not measured delivered throughput; delay is the queue divided by capacity proxy. Energy is the declared engineering proxy. No sampled outage or buffer overflow occurs; successful original reward arrival retains 123.65 kbit, which the shared requirements permit. Connecting sampled points does not establish connectivity between samples.

TABLE IV

V2.1 MINUS FULL V2 ON COMMON SUCCESSFUL FRESH PAIRS. VALUES ARE MEANS; INTERVALS DESCRIBE PAIRED DIFFERENCES. LOWER TIME, DELAY, HANDOVERS AND CONSUMED ENERGY AND HIGHER SINR ARE PREFERABLE.

Metric	Standard			Longer		
	Full V2	V2.1	95% difference interval	Full V2	V2.1	95% difference interval
Flight time (s)	28.347	28.594	[0.020, 0.520]	60.409	60.785	[0.039, 0.747]
Delay proxy (s)	4.742	1.291	[-3.957, -2.962]	6.199	1.583	[-4.957, -4.287]
Handovers	0.402	9.839	[8.968, 9.897]	1.442	23.058	[20.923, 22.287]
Consumed energy (kJ)	7.873	7.868	[-0.014, 0.005]	19.057	19.032	[-0.049, 0.002]
SINR (dB)	-8.916	-7.457	[1.296, 1.628]	-9.745	-7.978	[1.662, 1.870]
Interference (μ W)	0.736	0.725	[-0.019, -0.002]	0.763	0.755	[-0.016, -0.002]
Fixed V2 radio cost	5.174	6.923	[1.520, 1.983]	13.437	16.269	[2.377, 3.280]

TABLE V

V2.1 FOLLOWUP JOINT SUCCESS ON NEW 54012/54013 ROUTES. LEARNED ARMS USE 2,500 EPISODES PER SPLIT; REFERENCES USE 500. HIGHER IS PREFERABLE.

Controller	Standard (%)	Longer (%)
V2.1 service	95.48	88.60
Full V2	96.92	91.84
Original reward V1.5	96.04	89.48
Goal radio	97.40	91.80
Joint one step	96.20	91.40
Joint three step	94.00	92.80

$$C_t^{\text{svc}} = 0.10 \log(1 + D_t/(1s)) + 0.20(q_t/Q)^2 + 0.03H_t + 0.30 \frac{10^5 I_t}{1 + 10^5 I_t}. \quad (5)$$

It replaces only the training radio cost; V2 potential shaping, time cost and terminal payments remain. One second is a normalization scale, not a packet deadline. The selected candidate has no RSS shaping term. Transitions, observations, actions, navigation aid, filter, PPO and mission requirements remain identical. The filter and all reported fixed radio costs continue to use the original V2 cost, preventing a hidden controller change.

After selection, the source and coefficients were frozen. Five new policies used seeds 2101–2105 and 524,288 interactions each. They were compared with the five fixed V1.5 and five full V2 checkpoints on 500 new standard and 500 new longer routes, seeds 54012/54013. Goal radio and both joint search references were also evaluated. All 18,000 episodes replayed exactly. These routes are disjoint from prior saved routes but remain on the same map and within training distances.

The primary contrast is V2.1 minus full V2 standard success. A 95% crossed seed/route interval uses 5,000 draws and seed 64000. A favorable completion claim requires its lower bound above zero; a claim of completion improvement without a delay regression additionally requires the paired delay interval upper bound at most zero. Other intervals are descriptive. No tuning followed the fresh tests.

B. Fresh Results and Tradeoffs

Standard V2.1 minus full V2 is -1.440 percentage points, interval [-3.160, 0.320]. The fresh standard comparison does not establish improved completion over full V2. On longer routes the difference is -3.240 points, interval [-5.640, -0.880]. The longer route interval supports lower completion than full V2.

The paired service comparison contains 2335 standard and 2086 longer common successes. The standard paired delay difference is -3.451 s and the handover difference is +9.437, conditional on common successes. A service improvement does not establish improved completion or simultaneous improvement in every radio metric. All failures remain in the success denominators.

For standard V2.1 / full V2, first RSS failures are 111 / 71 and first buffer failures are 2 / 6; for longer routes they are 276 / 160 and 4 / 39, respectively. Other failure types and every seed are retained in note 48.

Against V1.5, standard completion differs by -0.560 points, interval [-2.040, 0.760]. This is a comparison with the written original reward on V2, not the original thesis agent. Selection from a small validation set and only five final training seeds limit inference.

VII. THESIS METRIC RECOVERY AND GUARDED CONTROL

We added passive SNR, all neighbor RSS power, remaining energy and handover frequency measurements. All 4,000 initial V1.5/full V2 episodes replayed with unchanged original fields. These added measurements are not independent new flights. Source Eq. (8) gives $\text{SNR} = \text{RSS} + 112.41 - 9 \text{ dB}$ [1, p. 10, Eq. (8); p. 15, Table 3]. The same map has maximum Operator 1 RSS of -25 dBm, so this expression cannot exceed 78.41 dB. The source plotted medians near 122–127 dB [1, pp. 18, 20, Figs. 6, 8] cannot follow from these stated inputs. We do not infer the coding cause. The source explicitly converts an initial noise value of -174 to -112.41 dBm [1, p. 14, Sec. 6.1; p. 15, Table 3]. If that conversion were omitted in the plotted calculation, subtracting 61.59 dB would give approximate median SNRs of 62.4/65.4 dB for equal PPO/greedy and 61.4/60.4/61.4 dB for delay/interference/handover policies. Note 57 records native image digitization and a conservative $\pm 1 \text{ dB}$ reading allowance. This is a conditional hypothesis,

TABLE VI
FRESH 55012/55013 JOINT SUCCESS (%). HIGHER IS BETTER. LEARNED
ARMS USE 2,500 FLIGHTS PER SPLIT; REFERENCES USE 500.

Controller	Standard	Longer
V1.5 original reward	93.80	89.12
Full V2	94.44	90.64
V2.1 service	93.16	88.64
V2.2 guarded V2	96.44	96.32
Goal radio	93.60	91.60
Joint one step	94.20	87.80
Joint three step	91.60	87.40

not a recovered actual result: source code and raw figure data remain unavailable. Neither this correction nor superiority over the source policies is established. For Eq. (9), we explicitly sum the linear RSS powers of all nonserving stations, without cochannel masking [1, p. 10, Eq. (9)]. This is a source equation proxy; it is not calibrated physical uplink. We retain cochannel downlink interference separately. Remaining energy is $100 - E_{\text{used}}/1000$ kJ under V2 accounting. The source recurrence adds electronics consumption, omits time, and labels capacity in kW [1, p. 10, Eq. (10); p. 15, Table 3]. Its literal score increases under the V2 speed cap and has no validated energy interpretation. Handovers use explicit counts and frequencies; source axis normalization remains unknown.

A. Declared Supervisor and Fresh Tests

A new supervisor considers the existing full V2 PPO motion and nine shared motion primitives. It simulates each followed by goal/braking guidance, chooses admissible network options by feasibility then RSS and capacity, and ranks rollouts by failure avoidance, arrival and remaining distance. The shared A3 rule and executed filter remain active. It adds computation and known map access, not a reward intervention or new training. Guard horizons 3, 5 and 8 completed 90, 92 and 92 of 96 validation missions with policy seed 2101; full V2 completed 91. The declared completion and delay rule selected horizon 5. Source and parameters were then frozen before generating 500 new standard and 500 new longer routes with seeds 55012/55013. The five full V2 policies were reused under the supervisor; V1.5, V2.1 and deterministic references were also evaluated. No tuning followed the fresh tests. The primary fresh comparison supports improved standard mission completion. V2.2 minus full V2 is +2.000 percentage points, 95 percent interval [0.320, 3.681], standard; longer is +5.680 [3.200, 8.280]. Intervals use 5,000 crossed seed/route bootstrap draws, seed 65000; secondary intervals are descriptive and unadjusted. Five reused policy seeds are not five new supervisor training runs. Service comparisons contain 2331 standard and 2225 longer common successful pairs. Every failed mission remains in completion denominators. Higher SNR and remaining energy and lower delay, interference and flight time are preferable; fewer handovers must preserve service. Notes 52–54 retain all source diagnostics, seed outcomes, failure reasons and reproduction commands. Delay decreases by 92.1 percent

and 92.5 percent on standard and longer common successes. SNR and SINR increase and both interference proxies decrease. Executed handovers increase, while energy and flight time differences versus full V2 have intervals containing zero. The guard meets the declared completion first and service priority in this simulator, at additional control and computation cost. Against the V1.5 control, it also improves completion and delay, but consumes more energy; complete secondary intervals are in note 53.

TABLE VII
FULL V2 / V2.2 MEANS ON COMMON SUCCESSFUL FRESH PAIRS. BRACKETS GIVE 95% INTERVALS FOR V2.2 MINUS FULL V2.

Metric and preferred direction	Standard means	Difference interval	Longer means	Difference interval
SNR (dB) $\uparrow$	62.238 / 64.667	[2.282, 2.576]	62.045 / 64.707	[2.563, 2.757]
All neighbor RSS power (μW) $\downarrow$	1.447 / 1.386	[-0.066, -0.057]	1.536 / 1.465	[-0.075, -0.068]
Remaining energy (kJ) $\uparrow$	91.766 / 91.762	[-0.012, 0.006]	80.620 / 80.607	[-0.024, 0.001]
Executed handovers $\downarrow$	0.466 / 4.811	[4.133, 4.564]	1.607 / 10.527	[8.584, 9.233]
Delay proxy (s) $\downarrow$	5.404 / 0.429	[-5.536, -4.441]	6.609 / 0.498	[-6.472, -5.744]
Flight time (s) $\downarrow$	29.384 / 29.392	[-0.079, 0.075]	61.330 / 61.288	[-0.157, 0.054]
SINR (dB) $\uparrow$	-9.355 / -5.775	[3.426, 3.732]	-9.828 / -6.035	[3.684, 3.896]
Cochannel interference (μW) $\downarrow$	0.741 / 0.566	[-0.182, -0.167]	0.775 / 0.594	[-0.186, -0.175]

Longer fresh routes: 500 routes $\times$ 5 fixed policy seeds
CDFs use 2225 common successful pairs; bars include all 2,500 flights per controller

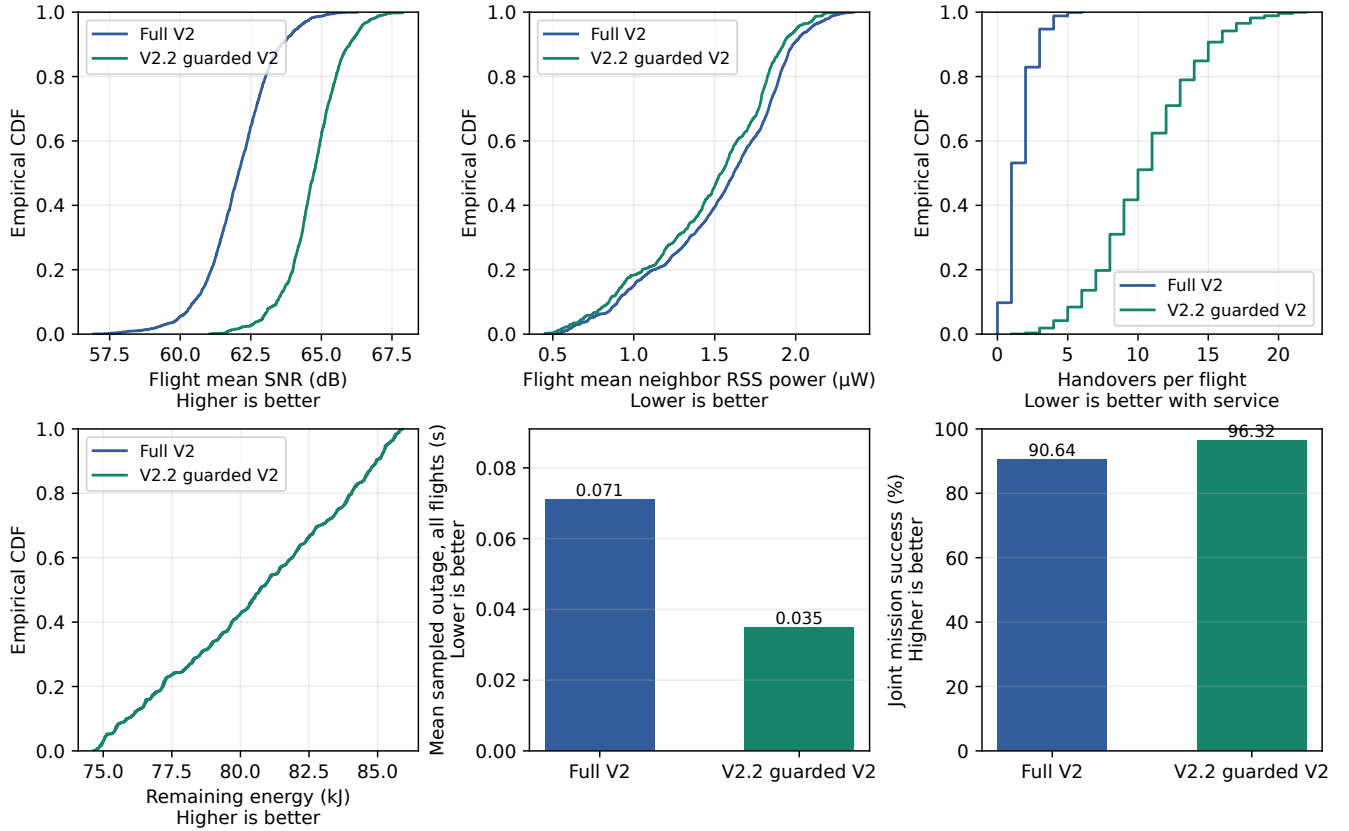


Fig. 4. Longer fresh route metrics. CDFs compare per flight values on common successes; bars include all flights. SNR is a per flight mean, all neighbor power is the explicit source equation proxy, and remaining energy uses V2 accounting. Sampled outage ends at the first violation and is censored. These are not reconstructed source figure distributions.

VIII. SIGNAL AWARE PATH SELECTION

TABLE VIII

FRESH V2.2 / V2.3 MEANS ON COMMON SUCCESSFUL PAIRS. BRACKETS GIVE 95% INTERVALS FOR V2.3 MINUS V2.2. HIGHER SNR AND LOWER OTHER QUANTITIES ARE PREFERABLE.

Metric	Standard	Difference interval	Longer	Difference interval
SNR (dB)	64.710 / 65.087	[0.340, 0.415]	64.696 / 64.979	[0.245, 0.327]
Delay (s)	0.377 / 0.277	[-0.120, -0.081]	0.492 / 0.396	[-0.119, -0.074]
Handovers	4.626 / 4.266	[-0.449, -0.278]	10.868 / 10.230	[-0.786, -0.494]
Flight time (s)	28.579 / 28.762	[0.132, 0.237]	61.419 / 61.634	[0.165, 0.266]
Consumed energy (kJ)	7.940 / 7.955	[0.010, 0.020]	19.437 / 19.453	[0.010, 0.021]

We next tested a separately declared path selection rule, V2.3, retaining V2.2's five step rollouts, ten motion candidates, feasible strongest signal association and executed filter. Outside 50 m of the goal, it chooses a higher predicted mean SNR path only within a fixed rollout progress allowance and with no predicted increase in delay or handovers. No weights, reward, RSS requirement or buffer limit change. SNR still follows the source Eq. (8) [1, p. 10, Eq. (8); p. 15, Table 3]. Development used 128 standard and 64 longer new routes (56001/56002), with fixed policies 2101 and 2102. Allowances 1, 3 and 6 m were tested, totaling 1536 episodes including V2.2. Only 1 m qualified. It preserved 252/256 standard and 122/128 longer completions while increasing matched mean SNR by 0.421/0.297 dB and reducing delay and handovers. Both larger allowances lost standard completions and exceeded the 2% standard time allowance; all candidates remain in note 59. After freezing source and selection, we generated 500 standard and 500 longer routes (56012/56013). Both controllers used all five unchanged full V2 policies, for 10000 final episodes. Every episode and sample array replayed exactly. No tuning followed the final tests. The 1 m allowance applies to predicted terminal progress within the same safety class, not total mission detour distance. Standard completion is 98.60% / 98.56% for V2.2/V2.3, difference -0.040 percentage points, 95% interval [-0.600, 0.520]. Longer completion is 93.72% / 93.60% for V2.2/V2.3, difference -0.120 percentage points, 95% interval [-0.880, 0.720]. The combined claim requires nonnegative completion differences with interval lower bounds above -1 percentage point in both splits; nonpositive delay and handover differences with upper bounds below 0.05 s and 0.25 switches; and time and energy increases below 2%. Subject to these gates, the SNR lower bound must exceed 0.1 dB standard and zero longer. These are declared study margins, not source requirements or proof of equal reliability. Intervals use 5000 crossed policy/route bootstrap draws, seed 66000. Other metrics are descriptive and unadjusted. There are 2459 standard and 2325 longer common successful pairs. Every failure remains in completion denominators. Note 59 reports SINR, interference, all failures, sample medians and every policy seed. V2.3 fails the combined gate and is not promoted over V2.2. Failed gates: test completion, longer test completion. Both completion intervals include zero and exclude a loss of one percentage point. The conservative nonnegative point requirement fails; a completion decrease is

not statistically resolved.

IX. INTERPRETATION AND LIMITATIONS

The contrast isolates the reward package conditional on V2. It does not identify individual terms or behavior without the navigation prior and prospective filter. These components may compensate for reward deficiencies. Removing them would require a separate experiment and fresh evaluation.

Original speed limits, motion, discrete action parameterization, continued training after arrival and unknown implementation details differ. Reconstructing the reward on V2 cannot identify the cause of the thesis's wandering [1, p. 21, Sec. 8]. The dataset is the same as the thesis's; sentinel meaning remains unverified. One map, static load, ideal service and uncalibrated energy limit inference. No physical continuity, packet latency, collision safety or deployment claim follows. A successful arrival can still retain queued data. The initial comparison uses seeds 53012/53013; the V2.1 followup uses fresh 54012/54013 routes; the supervisor study uses 55012/55013 and the signal path comparison uses 56012/56013. These test samples are reported separately. The supervisor has additional map queries and computation; its performance cannot be attributed to PPO training or reward replacement.

X. CONCLUSION

In the initial V1.5/full V2 comparison, reward replacement does not establish improved mission completion. Full V2 reduces handovers and flight time but increases delay and consumed energy. The separate V2.1 service reward improves delay while missing its completion first gate. Neither result justifies a reward only diagnosis of the original thesis's behavior. The separate V2.2 supervisor study supports improved standard mission completion. It improves delay and signal quality at the cost of more handovers than full V2. The additional metrics expose unresolved source arithmetic and aggregation; they do not establish physical uplink performance or a reproduced repair of the original agent. The further signal path rule has mean SNR differences of +0.377/+0.284 dB versus V2.2 on 10000 fresh episodes, and misses its declared combined gate. This result does not establish dominance on every metric or superiority over the unavailable thesis agent. The controller improvement is conditional on the shared simulator and additional lookahead computation. Further claims require independent maps, traffic and controller ablations rather than tuning on these test routes.

APPENDIX A ARTIFACTS AND REPRODUCTION

The repository includes all 15 initial comparison checkpoints and five V2.1 checkpoints, source, configuration, raw records, analysis, source snapshots and this paper: https://github.com/unworthyzeus/UAV_joint_handover_trajectory_optimization. The initial comparison uses ten unchanged V2 policies and five V1.5 policies; the reward followup adds five policies with the validation selected service reward. The supervisor and signal path selector add no weights and reuse the five full V2 policies. The private dataset belongs at `dataset/Barcelona_dataset_January.h5`; the setup guide records its checksum, dependency versions, exact commands and outputs. Note 36 is the initial frozen protocol, note 37 records its results, and note 38 records its delivery validation. Notes 46–49 document the V2.1 protocol, development, results and reproduction. Notes 51–54 cover the supervisor, recovered metrics, fresh results and model usage. Notes 58–59 document the signal path protocol, every candidate and policy, final gates and reproduction. The README documents the exact V2 differences from the TFM with page locators. This is a research draft, not an accepted publication.

APPENDIX B EVERY SEED AND OBSERVATION LAYOUT

TABLE IX
EVERY LEARNED SEED ON INITIAL 53012/53013 ROUTES: JOINT SUCCESS PERCENTAGES.

Arm	Seed	Standard	Longer
PPO original (V1.5)	2101	96.5	85.5
PPO original (V1.5)	2102	98.0	89.0
PPO original (V1.5)	2103	95.0	85.0
PPO original (V1.5)	2104	94.0	85.5
PPO original (V1.5)	2105	95.0	84.0
PPO full (V2)	2101	97.0	86.5
PPO full (V2)	2102	95.0	87.5
PPO full (V2)	2103	95.5	85.5
PPO full (V2)	2104	95.5	87.5
PPO full (V2)	2105	95.5	89.5
PPO arrival (V2)	2101	95.5	83.5
PPO arrival (V2)	2102	96.5	84.5
PPO arrival (V2)	2103	94.0	86.5
PPO arrival (V2)	2104	93.5	82.5
PPO arrival (V2)	2105	95.0	88.5

TABLE X
EVERY LEARNED SEED ON FRESH 54012/54013 ROUTES. ENTRIES ARE STANDARD / LONGER JOINT SUCCESS PERCENTAGES.

Seed	V1.5	Full V2	V2.1
2101	95.2 / 89.2	96.8 / 91.2	95.0 / 87.2
2102	96.6 / 89.2	97.2 / 91.2	96.4 / 89.0
2103	97.0 / 89.8	97.0 / 92.6	95.4 / 88.4
2104	95.4 / 89.0	96.6 / 91.4	96.0 / 88.4
2105	96.0 / 90.2	97.0 / 92.8	94.6 / 90.0

TABLE XI
EVERY FIXED POLICY SEED ON FRESH 55012/55013 ROUTES. ENTRIES ARE STANDARD / LONGER SUCCESS PERCENTAGES; HIGHER IS BETTER. V2.2 REUSES FULL V2 WEIGHTS.

Seed	V1.5	Full V2	V2.1	V2.2
2101	94.4 / 88.2	95.0 / 91.2	92.0 / 89.8	96.6 / 96.4
2102	95.0 / 90.2	94.6 / 91.8	94.0 / 88.2	96.4 / 96.4
2103	93.2 / 89.0	94.6 / 92.8	93.4 / 89.8	96.4 / 96.4
2104	93.2 / 88.4	94.6 / 88.0	93.8 / 88.2	96.4 / 96.0
2105	93.2 / 89.8	93.4 / 89.4	92.6 / 87.2	96.4 / 96.4

TABLE XII
EVERY FIXED POLICY SEED ON FRESH 56012/56013 ROUTES. ENTRIES ARE SUCCESSFUL FLIGHTS OF 500; HIGHER IS BETTER.

Seed	Standard V2.2 / V2.3	Longer V2.2 / V2.3
2101	493 / 493	471 / 469
2102	493 / 492	467 / 469
2103	493 / 494	468 / 467
2104	493 / 492	469 / 467
2105	493 / 493	468 / 468

TABLE XIII
IDENTICAL V2 OBSERVATION LAYOUT IN ALL FOUR LEARNED REWARD ARMS, INCLUDING V2.1.

Indices	Meaning
0–1	Position / map extent
2–3	Unit goal direction
4	Goal distance / 1,000 m
5–6	Radial/lateral velocity / 25 m/s
7–8	Speed / 25; elapsed steps / 200
9–10	Queue and consumed energy fractions
11–13	Load phase, RBG and serving SINR
14–18	Candidate RSS values
19–28	Candidate relative coordinates
29–33	Candidate station indices
34–94	Log capacity for 61 options
95–155	Network mask

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